# Latency Controllers for Real-Time Audio Streaming over IP

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**Abstract**

Vibe-mathed mathematical setting for latency control: with optimality criteria (latency bias, latency variance, warp factor smoothness), an SDE model of jitter, packet arrivals and sample queue depth, and derived optimal controllers.

## Contents

# 1 Introduction

TODO

# 2 System Model

We consider a sender and a receiver, each equipped with an independent free-running clock. Neither clock is assumed to be "correct"; instead we introduce an absolute reference time $t \in \mathbb{R}$ that is external to both devices (one may think of it as an ideal UTC clock that neither party has access to).

## 2.1 Clocks and Drift

Let $\tau_A(t)$ denote the reading of the *sender clock* (clock A) at absolute time $t$, and $\tau_B(t)$ the reading of the *receiver clock* (clock B). Each clock runs at a rate that deviates from the reference by a small, slowly-varying fractional offset:

$$\frac{\mathrm{d}\tau_A}{\mathrm{d}t} = 1 + \alpha(t), \tag{1}$$

$$\frac{\mathrm{d}\tau_B}{\mathrm{d}t} = 1 + \beta(t), \tag{2}$$

where $\alpha(t)$ and $\beta(t)$ are the instantaneous fractional frequency offsets of clocks A and B respectively (typically on the order of $10^{-6}$–$10^{-5}$, i.e. a few ppm for quartz oscillators).

**Assumption 2.1** (Nearly-stationary drift)**.** The offsets $\alpha$ and $\beta$ are approximately constant over the controller's timescale (seconds to minutes) but may drift slowly over longer horizons (tens of minutes to hours) due to temperature or voltage changes. Formally, we treat each as a piecewise-linear function of $t$ whose slope is small enough to be neglected within any single control epoch.

Under Assumption 2.1, integrating (1) gives

$$\tau_A(t) \;=\; t + \int_0^t \alpha(s)\,\mathrm{d}s \;=\; (1+\alpha)\,t, \tag{3}$$

where the equality holds when $\alpha$ is treated as constant. An analogous expression holds for $\tau_B$.

The quantity that ultimately matters for clock synchronisation is the *relative drift* between sender and receiver:

$$\delta \;=\; \frac{(1+\alpha) - (1+\beta)}{1+\beta} \;=\; \frac{\alpha - \beta}{1+\beta}. \tag{4}$$

*Remark* 2.1. The receiver never observes $\alpha$ or $\beta$ individually—only their difference $\delta$ is identifiable from packet timing. The reference time $t$ is a modelling convenience; all observable quantities will be expressed in receiver-clock time $\tau_B$.

## 2.2 Packet Emission

The sender packetises audio into frames of $N$ samples at nominal sample rate $F_s$ (Hz). Packet $n$ ($n = 0, 1, 2, \dots$) is emitted when the sender clock reads

$$\tau_A^{(\mathrm{send})}(n) \;=\; n\,\Delta, \qquad \Delta \;=\; \frac{N}{F_s}, \tag{5}$$

where $\Delta$ is the *packet period* (seconds of sender-clock time).

The corresponding absolute time of emission is obtained by inverting (3):

$$t^{(\mathrm{send})}(n) \;=\; \frac{n\,\Delta}{1+\alpha}. \tag{6}$$

### 2.3 Network Transit and Arrival Times

Packet $n$ traverses the network and arrives at the receiver at absolute time

$$t^{(\mathrm{arr})}(n) \;=\; t^{(\mathrm{send})}(n) + d(n), \tag{7}$$

where $d(n)$ is the *one-way network delay* for packet $n$.

We assume that FEC fully recovers all losses, so every emitted packet eventually arrives. The delay is decomposed as

$$d(n) \;=\; d_0 + \eta(n), \tag{8}$$

where $d_0 > 0$ is a baseline (minimum or mean) propagation delay and $\eta(n)$ is the *jitter*—the stochastic, packet-varying component of delay.

**Assumption 2.2** (Jitter model)**.** The jitter process $\eta(n)$ may exhibit short-range correlations (e.g. bursty congestion, queuing effects), but is *long-range uncorrelated* in the following sense: there exists a mixing horizon $M$ such that for $|i-j| > M$, $\eta(i)$ and $\eta(j)$ are approximately independent.

On a per-packet timescale we define the jitter increments

$$\xi(n) \;=\; \eta(n) - \eta(n-1), \tag{9}$$

and require only that they are zero-mean with finite variance:

$$\mathbb{E}[\xi(n)] = 0, \qquad \mathrm{Var}[\xi(n)] = \sigma_\xi^2 < \infty. \tag{10}$$

No further distributional assumptions are imposed.

*Remark* 2.2 (Packet reordering). This model admits packet reordering: if $\xi(n) < -\Delta/(1+\alpha)$, then $t^{(\mathrm{arr})}(n) < t^{(\mathrm{arr})}(n-1)$, i.e. packet $n$ arrives before packet $n-1$. The jitter buffer at the receiver is responsible for restoring packet order before playback.

**Arrival time in receiver clock.** The receiver observes arrivals in its own clock $\tau_B$. Applying (2) to (7):

$$\tau_B^{(\mathrm{arr})}(n) \;=\; (1+\beta)\,t^{(\mathrm{arr})}(n) \;=\; (1+\beta)\left[\frac{n\,\Delta}{1+\alpha} + d_0 + \eta(n)\right]. \tag{11}$$

For the *inter-arrival time* observed by the receiver:

$$\tau_B^{(\mathrm{arr})}(n) - \tau_B^{(\mathrm{arr})}(n-1) \;=\; (1+\beta)\left[\frac{\Delta}{1+\alpha} + \xi(n)\right] \;=\; \frac{\Delta}{1+\delta} \;+\; (1+\beta)\,\xi(n), \tag{12}$$

where the second equality uses $\frac{1+\beta}{1+\alpha} = \frac{1}{1+\delta}$ from (4).

This is the fundamental measurement equation: each inter-arrival interval reveals the packet period $\Delta$ scaled by the relative drift $1/(1+\delta)$, plus a jitter noise term scaled by $(1+\beta)$.

## 3 Controller Model

All quantities in this section are expressed in receiver-clock time, which we denote $\tau \equiv \tau_B$ for brevity.

### 3.1 Arrival Process

The *cumulative arrival process* is

$$A(\tau) \;=\; N \cdot \#\big\{\, n \ge 0 \;:\; \tau_B^{(\mathrm{arr})}(n) \le \tau \,\big\}, \tag{13}$$

the total number of audio samples deposited into the jitter buffer by receiver-clock time $\tau$. This is a right-continuous, non-decreasing, pure-jump process with jumps of size $N$ at each packet arrival.

### 3.2 Warp Rate

**Definition 3.1** (Warp rate)**.** The *warp rate* $u(\tau) \in \mathbb{R}$ is the fractional adjustment to the nominal playback rate. The receiver's *desired consumption rate* of audio samples is

$$r(\tau) \;=\; F_s\big(1 + u(\tau)\big) \tag{14}$$

samples per unit of receiver-clock time. When $u = 0$, playback proceeds at the nominal sample rate $F_s$. In the implementation, the quantity $1 + u(\tau)$ corresponds to `freq_coeff`.

### 3.3 Queue Dynamics

The jitter buffer holds samples that have arrived but have not yet been played back. Samples are added in bursts of $N$ at each packet arrival (via $A$) and consumed continuously at the desired rate $r(\tau)$, subject to the constraint that the buffer cannot go negative: when the buffer is empty, silence is inserted instead of audio.

**Definition 3.2** (Queue depth and underrun regulator)**.** The *queue depth* $q(\tau)$ and the *underrun regulator* $L(\tau)$ are the unique pair of right-continuous processes satisfying the following system for all $\tau \ge 0$:

$$q(\tau) \;=\; q(0) \;+\; A(\tau) \;-\; \int_0^{\tau} F_s\big(1 + u(s)\big)\,\mathrm{d}s \;+\; L(\tau), \tag{15}$$

with the conditions

(i) **Non-negativity.** $q(\tau) \ge 0$ for all $\tau \ge 0$.

(ii) **Monotonicity.** $L$ is non-decreasing and $L(0) = 0$.

(iii) **Complementarity.** $L$ increases only when $q = 0$:

$$\int_0^{\infty} \mathbf{1}\{q(\tau) > 0\}\;\mathrm{d}L(\tau) \;=\; 0. \tag{16}$$

Equation (15) is a one-sided Skorokhod reflection of the *free queue process*

$$\tilde{q}(\tau) \;=\; q(0) + A(\tau) - \int_0^{\tau} F_s\big(1 + u(s)\big)\,\mathrm{d}s, \tag{17}$$

which is the queue depth that would result if playback were never interrupted by an empty buffer. The regulator $L$ is the minimal non-negative correction that keeps $q \ge 0$; it is given explicitly by

$$L(\tau) \;=\; \sup_{0 \le s \le \tau} \big(-\tilde{q}(s)\big)^{+}, \tag{18}$$

where $x^{+} = \max(x, 0)$.

**Definition 3.3** (Underrun)**.** An *underrun event* occurs at time $\tau$ when $q(\tau) = 0$ and the desired consumption rate $r(\tau) > 0$. During an underrun, the receiver outputs silence.

The regulator $L(\tau)$ has a direct physical interpretation: it is the cumulative number of *silence samples* inserted up to time $\tau$ in place of audio that was not available. The *underrun time* is

$$U(\tau) \;=\; \int_0^\tau \mathbf{1}\{q(s) = 0\}\, \mathrm{d}s, \tag{19}$$

the total duration for which the buffer has been empty.

*Remark* 3.1 (Relationship between $L$ and $U$). The processes $L$ and $U$ are related but distinct. The underrun time $U$ measures duration; the regulator $L$ measures samples. By the complementarity condition, $L$ increases only on the set $\{q = 0\}$, so

$$L(\tau) \;=\; \int_0^\tau F_s\big(1 + u(s)\big)\, \mathbf{1}\{q(s) = 0\}\, \mathrm{d}s. \tag{20}$$

**Definition 3.4** (Overrun and session restart)**.** The system enforces an upper bound $Q_{\max}$ on the queue depth. The *overrun stopping time* is

$$T_{\text{over}} \;=\; \inf\big\{\, \tau \geq 0 \;:\; q(\tau) > Q_{\max} \,\big\}, \tag{21}$$

with $T_{\text{over}} = \infty$ if the bound is never exceeded. When $T_{\text{over}} < \infty$, the receiver terminates the current session and initiates a *session restart*: the buffer is flushed, all estimator state is reset, and a new session begins with fresh initial conditions.

The queue dynamics (15)–(16) and the controller (22) are therefore only defined on $[0, T_{\text{over}})$. A well-designed controller should ensure $T_{\text{over}} = \infty$ almost surely under normal operating conditions.

### 3.4 Observable Information

The receiver does not have access to the absolute reference time $t$, the sender clock $\tau_A$, or the drift parameters $\alpha$, $\beta$, $\delta$ individually.

At receiver-clock time $\tau$, the following quantities are observable:

(a) The queue depth $q(s)$ for $s \leq \tau$.

(b) The arrival times $\tau_B^{(\text{arr})}(n)$ for all packets received by time $\tau$.

(c) The RTP timestamps carried by each received packet, which encode the sender-clock emission times $\tau_A^{(\text{send})}(n) = n\,\Delta$.

We collect these into the *observable filtration* $\{\mathcal{F}_\tau\}_{\tau \geq 0}$, the $\sigma$-algebra generated by the history of (a)–(c) up to time $\tau$.

### 3.5 Definition of a Controller

**Definition 3.5** (Controller)**.** A *controller* is a causal (non-anticipative) functional

$$u(\tau) \;=\; \Gamma\big(\mathcal{F}_\tau\big), \tag{22}$$

mapping the observable history up to time $\tau$ to a warp rate $u(\tau) \in \mathbb{R}$. The closed-loop system is the coupled solution of (15)–(16) and (22).

# 4 Objectives

A controller $\Gamma$ induces a closed-loop trajectory $(q, u, L, U)$ on $[0, T_{\text{over}})$. We now formalise the criteria by which controllers are compared.

## 4.1 Warp Smoothness

Abrupt changes in the warp rate $u$ produce audible pitch transients. A constant warp rate (matching the true drift) is inaudible; only its *variation* causes artefacts. We therefore penalise the temporal roughness of $u$.

**Definition 4.1** (Warp cost)**.** For a controller producing warp trajectory $u$ on $[0, T]$, the *warp cost* is the $\dot{H}^1$ seminorm

$$J_u(T) \;=\; \int_0^T \left| \frac{\mathrm{d}u}{\mathrm{d}\tau}(\tau) \right|^2 \mathrm{d}\tau. \tag{23}$$

A trajectory with $J_u = 0$ corresponds to a constant warp rate.

*Remark* 4.1. The $\dot{H}^1$ seminorm penalises only the derivative of $u$, not its magnitude. This is deliberate: the steady-state value of $u$ is dictated by the clock drift $\delta$ and is not a design choice. What the controller *can* control is how smoothly it converges to that value.

## 4.2 Latency Regulation

Let $q^*$ denote the *target queue depth* (in samples) and define the *queue error*

$$e(\tau) \;=\; q(\tau) - q^*. \tag{24}$$

We decompose the latency performance into two components.

**Bias (mean error).** The long-run time-averaged queue error is

$$B(T) \;=\; \frac{1}{T} \int_0^T e(\tau)\; \mathrm{d}\tau. \tag{25}$$

A well-tuned controller should drive $B(T) \to 0$ as $T \to \infty$.

**Variance (error fluctuation).** The time-averaged mean-square error is

$$V(T) \;=\; \frac{1}{T} \int_0^T e(\tau)^2\; \mathrm{d}\tau. \tag{26}$$

This captures both persistent bias and stochastic fluctuation. In the ergodic regime where $B(T) \to 0$, $V(T)$ converges to the stationary variance of the queue error.

*Remark* 4.2 (Multi-speaker synchronisation). In multi-speaker setups, several receivers play the same audio stream. Each receiver $i$ runs its own controller and produces its own queue trajectory $q_i(\tau)$ with error $e_i(\tau)$. Phase coherence across speakers requires that the *inter-speaker* variance

$$V_{\text{sync}}(\tau) \;=\; \frac{1}{K} \sum_{i=1}^{K} \big(q_i(\tau) - \bar{q}(\tau)\big)^2, \qquad \bar{q}(\tau) = \frac{1}{K} \sum_{i=1}^{K} q_i(\tau), \tag{27}$$

be small. Since each $q_i$ fluctuates independently around its target, minimising the per-receiver variance $V$ directly minimises $V_{\text{sync}}$.

### 4.3 Underrun and Overrun Avoidance

#### 4.3.1 Overruns

An overrun triggers a session restart (Def. 3.4), which is a catastrophic event: all accumulated state is lost and the stream audibly glitches. Since the process terminates at $T_{\text{over}}$, overrun avoidance is naturally expressed via the stopping time.

(O1) **Survival probability.** For a given time horizon $T$,

$$P\big(T_{\text{over}} > T\big) \;\to\; 1. \tag{28}$$

(O2) **Expected lifetime.** $\mathbb{E}[T_{\text{over}}] \to \infty$, or more precisely, requiring that all moments $\mathbb{E}[T_{\text{over}}^k]$ grow appropriately.

A well-designed controller should ensure $T_{\text{over}} = \infty$ a.s. under the stationary drift and jitter assumptions (Assumptions 2.1 and 2.2).

#### 4.3.2 Underruns

Unlike overruns, an underrun does not terminate the process—the receiver inserts silence and continues. A good controller should recover from underruns and prevent them from recurring. This suggests metrics that measure the *density* of underruns rather than a single stopping time.

(U1) **Underrun rate (samples).** The long-run average rate of silence insertion:

$$R_L \;=\; \limsup_{T\to\infty} \frac{L(T)}{T}. \tag{29}$$

This has units of samples per unit time and should be driven to zero.

(U2) **Underrun fraction (time).** The fraction of time spent in underrun:

$$R_U \;=\; \limsup_{T\to\infty} \frac{U(T)}{T}. \tag{30}$$

This is the long-run probability that the buffer is empty at a uniformly chosen instant.

(U3) **Underrun event rate.** Let $N_{\text{under}}(T)$ count the number of transitions $q(\tau) : {>}0 \;\to\; 0$ on $[0, T]$. The event rate

$$R_N \;=\; \limsup_{T\to\infty} \frac{N_{\text{under}}(T)}{T} \tag{31}$$

measures how frequently underruns occur, regardless of their duration.

*Remark* 4.3. These three metrics are not equivalent. A single long underrun contributes heavily to $R_L$ and $R_U$ but only once to $R_N$. Conversely, many brief underruns inflate $R_N$ while contributing little to $R_U$. In practice, all three should be small; the choice of which to optimise depends on the application's sensitivity to silence gaps versus playback interruptions.

## 5 Fluid Approximation

The exact queue dynamics (Definition 3.2) involve a pure-jump arrival process, making direct transfer-function analysis difficult. In this section we replace the discrete packet arrivals with a continuous flow and derive an Itô stochastic differential equation for the queue error. All quantities remain in receiver-clock time $\tau$.

## 5.1 Discrete Queue Recurrence

We first derive the exact per-packet queue dynamics. Let

$$q_n \;\coloneqq\; q\big(\tau_B^{(\mathrm{arr})}(n)^+\big) \tag{32}$$

denote the queue depth immediately after the $n$-th packet's $N$ samples have been deposited. Between arrivals $n-1$ and $n$, the receiver consumes samples at rate $F_s(1+u)$ for the inter-arrival duration

$$\Delta\tau(n) \;\coloneqq\; \tau_B^{(\mathrm{arr})}(n) - \tau_B^{(\mathrm{arr})}(n-1) \;=\; \frac{\Delta}{1+\delta} + (1+\beta)\,\xi(n) \tag{33}$$

(from (12)). Ignoring the reflecting barrier (i.e. assuming $q > 0$ throughout), the recurrence is

$$q_n \;=\; q_{n-1} \;+\; N \;-\; F_s\big(1+u_n\big)\,\Delta\tau(n), \tag{34}$$

where $u_n$ is the warp rate in effect during the interval (assumed constant over one packet period).

**Expanding.** Substituting $N = F_s\,\Delta$ and (33) into (34):

$$\begin{aligned} q_n - q_{n-1} &= F_s\,\Delta \;-\; F_s(1+u_n)\left[\frac{\Delta}{1+\delta} + (1+\beta)\,\xi(n)\right] \\ &= F_s\,\Delta\left[1 - \frac{1+u_n}{1+\delta}\right] \;-\; F_s(1+u_n)(1+\beta)\,\xi(n). \end{aligned} \tag{35}$$

The first term is the deterministic drift. Since $1 - \frac{1+u}{1+\delta} = \frac{\delta-u}{1+\delta}$, it equals

$$F_s\,\Delta\,\frac{\delta-u_n}{1+\delta}. \tag{36}$$

The second term is the stochastic jitter contribution:

$$-\,F_s(1+u_n)(1+\beta)\,\xi(n). \tag{37}$$

## 5.2 Summing Over a Control Epoch

Consider a control epoch of receiver-clock duration $h$, during which the warp rate $u$ is held constant. The number of packets arriving in this epoch is

$$M(h) \;=\; \#\big\{\,n : \tau_B^{(\mathrm{arr})}(n) \in [\tau,\,\tau+h]\,\big\}. \tag{38}$$

Since $\mathbb{E}[\Delta\tau(n)] = \Delta/(1+\delta)$, the expected count is

$$\mathbb{E}[M(h)] \;=\; \frac{h(1+\delta)}{\Delta}. \tag{39}$$

Summing (35) over the $M(h)$ packets in the epoch:

$$q(\tau+h) - q(\tau) \;=\; F_s(\delta-u)\,h \;-\; F_s(1+u)(1+\beta)\sum_{n\in\mathrm{epoch}}\xi(n), \tag{40}$$

where we have used $M(h)\cdot\frac{F_s\,\Delta}{1+\delta} = F_s\,h$ for the deterministic term (replacing $M$ by its mean; the fluctuation in $M$ is itself a jitter effect already captured by the $\xi$ sum).

### 5.3 Central Limit Passage

**Assumption 5.1** (Timescale separation)**.** The control update period $h$ satisfies $h \gg \Delta$ and $h \gg M\,\Delta$, where $M$ is the mixing horizon from Assumption 2.2. That is, many packets arrive per control epoch, and the epoch spans many mixing lengths.

Under Assumption 5.1, the epoch contains $M(h) \gg 1$ jitter increments $\xi(n)$, many of which are approximately independent (by the mixing condition in Assumption 2.2). By Donsker's invariance principle for mixing sequences, the partial-sum process

$$S(m) \;=\; \sum_{n=1}^{m} \xi(n) \tag{41}$$

converges (in distribution, under diffusive rescaling) to a Brownian motion:

$$\frac{1}{\sqrt{m}}\, S(\lfloor m\,\cdot \rfloor) \;\overset{d}{\Longrightarrow}\; \sigma_{\text{eff}}\, W(\cdot), \qquad m \to \infty, \tag{42}$$

where $W$ is a standard Wiener process and the *effective jitter standard deviation* is

$$\sigma_{\text{eff}}^2 \;=\; \sum_{k=-\infty}^{\infty} \operatorname{Cov}\big(\xi(0),\, \xi(k)\big) \;=\; \sigma_\xi^2 + 2\sum_{k=1}^{\infty} \operatorname{Cov}\big(\xi(0),\, \xi(k)\big). \tag{43}$$

This is the spectral density of the $\xi$ process evaluated at frequency zero. If $\xi$ is uncorrelated (i.i.d.), then $\sigma_{\text{eff}} = \sigma_\xi$. Short-range positive correlations increase $\sigma_{\text{eff}}$; negative correlations decrease it.

*Remark* 5.1. The convergence $\sigma_{\text{eff}}^2 < \infty$ is guaranteed by the mixing condition in Assumption 2.2: the autocovariances decay fast enough for the series to converge absolutely.

### 5.4 The Fluid SDE

Applying the CLT result to (40), the queue depth $q(\tau)$ satisfies the Itô stochastic differential equation

$$\boxed{\mathrm{d}q \;=\; F_s(\delta - u)\,\mathrm{d}\tau \;+\; \sigma_w\,\mathrm{d}W(\tau),} \tag{44}$$

where $W$ is a standard Wiener process and the *diffusion coefficient* is

$$\sigma_w \;=\; F_s\,(1+u)(1+\beta)\,\sigma_{\text{eff}}\sqrt{\frac{1+\delta}{\Delta}}. \tag{45}$$

**Derivation of $\sigma_w$.** In one epoch of duration $h$, the noise term in (40) has variance

$$\operatorname{Var}\left[F_s(1+u)(1+\beta)\sum_{n=1}^{M(h)} \xi(n)\right] \;=\; F_s^2(1+u)^2(1+\beta)^2 \;\cdot\; M(h)\,\sigma_{\text{eff}}^2. \tag{46}$$

Substituting $M(h) = h(1+\delta)/\Delta$ from (39), the variance is proportional to $h$:

$$\sigma_w^2\, h \;=\; F_s^2(1+u)^2(1+\beta)^2\,\frac{(1+\delta)}{\Delta}\,\sigma_{\text{eff}}^2\, h, \tag{47}$$

confirming the diffusive scaling and yielding (45).

**Simplified form.** Since $|u|, |\beta|, |\delta| \ll 1$ (all at the ppm level), to leading order

$$\sigma_w \;=\; \frac{F_s\,\sigma_{\text{eff}}}{\sqrt{\Delta}} \;=\; \frac{F_s^{3/2}\,\sigma_{\text{eff}}}{\sqrt{N}}. \tag{48}$$

We use this simplified form in subsequent analysis.

*Remark* 5.2. The second equality follows from $\Delta = N/F_s$. The dependence $\sigma_w \propto 1/\sqrt{N}$ is intuitive: larger packets (more samples per packet) mean fewer arrivals per unit time, so each jitter increment has more influence per unit of consumed time.

### 5.5 Queue Error SDE

In terms of the queue error $e(\tau) = q(\tau) - q^*$ (eq. (24)), the fluid SDE becomes

$$\mathrm{d}e \;=\; F_s(\delta - u)\,\mathrm{d}\tau \;+\; \sigma_w\,\mathrm{d}W(\tau), \tag{49}$$

since $q^*$ is constant. This is valid on the region $\{q > 0\} \cap \{q < Q_{\max}\}$, i.e. away from the reflecting barrier and the overrun boundary.

### 5.6 Validity and Limitations

The fluid approximation (44) is accurate when:

(a) The control timescale is much longer than the packet period $\Delta$ (Assumption 5.1).

(b) The queue depth $q$ is well above zero, so the reflecting barrier is not active. Near $q = 0$, the discrete nature of arrivals matters: an underrun lasts until the next packet arrives (a random time on the order of $\Delta$), which the continuous model cannot capture.

(c) The jitter increments $\xi(n)$ have finite $\sigma_{\text{eff}}$ (eq. (43)). Heavy-tailed jitter (infinite variance) would invalidate the CLT and require a different scaling limit (e.g. $\alpha$-stable processes).

## 6 Analysis of the Proportional Controller

We now analyse the simplest non-trivial controller under the fluid model: feedforward drift cancellation plus proportional feedback on the queue error. All results in this section follow from elementary properties of the Ornstein–Uhlenbeck process.

### 6.1 Pure Feedforward is Insufficient

Suppose the controller has perfect knowledge of the relative drift $\delta$ and sets $u(\tau) = \delta$ for all $\tau$ (pure feedforward, no feedback). The error SDE (49) becomes

$$\mathrm{d}e \;=\; \sigma_w\,\mathrm{d}W(\tau), \tag{50}$$

a driftless Brownian motion. Its variance grows without bound:

$$\mathrm{Var}[e(\tau)] \;=\; \sigma_w^2\,\tau. \tag{51}$$

By the recurrence of one-dimensional Brownian motion, the process hits every level in finite time almost surely. In particular, both the underrun barrier ($e = -q^*$) and the overrun barrier ($e = Q_{\max} - q^*$) are reached in finite expected time:

$$\mathbb{E}[T_{\text{hit}}] \;=\; \frac{q^*\,(Q_{\max} - q^*)}{\sigma_w^2}. \tag{52}$$

No amount of buffering can prevent eventual session death under pure feedforward control.

### 6.2 Feedforward Plus Proportional Feedback

Now consider the controller

$$u(\tau) \;=\; \delta + K\, e(\tau), \tag{53}$$

where $K > 0$ is the proportional gain. Substituting into (49):

$$\mathrm{d}e \;=\; -F_s\, K\, e\; \mathrm{d}\tau \;+\; \sigma_w\, \mathrm{d}W(\tau). \tag{54}$$

This is the *Ornstein–Uhlenbeck* (OU) SDE with mean-reversion rate $\theta = F_s\, K$ and diffusion coefficient $\sigma_w$.

### 6.3 Stationary Distribution

The OU process (54) has a unique stationary distribution:

$$e \;\sim\; \mathcal{N}\!\left(0,\; \frac{\sigma_w^2}{2\, F_s\, K}\right) \qquad \text{(in steady state)}. \tag{55}$$

Unlike the pure feedforward case, the variance is bounded and time-independent.

### 6.4 The Fundamental Tradeoff

The controller gain $K$ simultaneously determines the distributions of the queue error $e$ and the warp rate $u$. Since $u = \delta + K\, e$ and $\delta$ is (approximately) constant, the stochastic part of $u$ is $K\, e$.

**Proposition 6.1** (Stationary variances)**.** *Under the proportional controller* (53) *with the fluid model* (49)*, the stationary variances of the queue error and the warp rate are*

$$\mathrm{Var}[e] \;=\; \frac{\sigma_w^2}{2\, F_s\, K}, \tag{56}$$

$$\mathrm{Var}[u] \;=\; K^2\; \mathrm{Var}[e] \;=\; \frac{K\, \sigma_w^2}{2\, F_s}. \tag{57}$$

These move in **opposite directions** as $K$ varies:

| | $K\uparrow$ | $K\downarrow$ |
|---|---|---|
| $\mathrm{Var}[e]$ (queue error) | $\downarrow$ | $\uparrow$ |
| $\mathrm{Var}[u]$ (warp variation) | $\uparrow$ | $\downarrow$ |

Eliminating $K$ between (56) and (57) yields the *Pareto frontier*:

$$\boxed{\mathrm{Var}[e] \;\cdot\; \mathrm{Var}[u] \;=\; \frac{\sigma_w^4}{4\, F_s^2}.} \tag{58}$$

This is a hyperbola in the ($\mathrm{Var}[e]$, $\mathrm{Var}[u]$) plane. No proportional controller can achieve a point below this curve: any reduction in queue error variance must be paid for by an increase in warp rate variance, and vice versa. The product is fixed by the jitter intensity $\sigma_w$ and the sample rate $F_s$.

*Remark* 6.1. The Pareto frontier (58) arises from the *proportional* controller family. A more general controller (e.g. one that also acts on $\mathrm{d}u/\mathrm{d}\tau$) could potentially achieve points below this curve, at the cost of introducing higher-order dynamics.

### 6.5 Underrun Probability

In steady state, the queue error $e$ is Gaussian with zero mean and variance $\mathrm{Var}[e]$ from (56). An underrun occurs when $e < -q^*$, i.e. $q < 0$. The steady-state probability of being in an underrun state is

$$P(q < 0) \;=\; P(e < -q^*) \;=\; \Phi\!\left(\frac{-q^*}{\sqrt{\mathrm{Var}[e]}}\right) \;=\; \Phi\!\left(-q^* \sqrt{\frac{2\,F_s\,K}{\sigma_w^2}}\right), \tag{59}$$

where $\Phi$ is the standard normal CDF. This decreases rapidly (Gaussian tail) as either $q^*$ or $K$ increases.

*Remark* 6.2. This is the steady-state occupancy probability $R_U$ from eq. (30). For practical targets (e.g. $q^* = 5\sqrt{\mathrm{Var}[e]}$), the underrun probability is negligibly small ($< 10^{-6}$). The tradeoff is that a larger $q^*$ means higher nominal latency.

## 7 Spectral Analysis

The results of section 6 were derived via the time-domain properties of the OU process. In this section we re-derive them in the frequency domain, which reveals a more general principle: for linear systems driven by Gaussian noise, the *transfer function completely determines the output probability distribution.*

### 7.1 Plant Transfer Function

The fluid error SDE (49), written in input–output form, is

$$\dot{e}(\tau) \;=\; -F_s\,u(\tau) \;+\; w(\tau), \tag{60}$$

where $w(\tau) = \sigma_w\,\dot{W}(\tau)$ is white Gaussian noise with (two-sided) power spectral density

$$S_w(\omega) \;=\; \sigma_w^2 \tag{61}$$

and we have absorbed the constant $F_s\,\delta$ into the feedforward part of $u$ (i.e. we take $u$ to include the drift cancellation, so the remaining dynamics are driven by noise alone).

Taking the Laplace transform of (60) with zero initial conditions:

$$s\,E(s) \;=\; -F_s\,U(s) \;+\; W(s). \tag{62}$$

The open-loop plant from $U$ to $E$ is the pure integrator

$$G(s) \;=\; \frac{E(s)}{W(s) - F_s\,U(s)} \;=\; \frac{1}{s}, \tag{63}$$

and the noise enters the error through the same integrator: $E(s) = \frac{1}{s}\big[W(s) - F_s\,U(s)\big]$.

*Remark* 7.1 (Intuition: the integrator). The plant $G(s) = 1/s$ is an integrator: it accumulates its input over time. In the frequency domain, this means that slow disturbances (low $\omega$) are amplified much more than fast ones (high $\omega$), since $|1/j\omega|$ is large when $\omega$ is small. Physically, a slow persistent jitter bias drains or fills the buffer steadily, while a fast oscillating jitter averages out before the queue can respond. This is why drift (a zero-frequency disturbance) is so dangerous: the integrator amplifies it to infinity.

## 7.2 PSD Propagation

For any linear time-invariant system with transfer function $H(s)$ driven by a wide-sense stationary input with PSD $S_{\text{in}}(\omega)$, the output PSD is

$$S_{\text{out}}(\omega) \;=\; \big|H(j\omega)\big|^2 \, S_{\text{in}}(\omega), \tag{64}$$

and the output variance (the total power) is

$$\text{Var}[\text{output}] \;=\; \int_{-\infty}^{\infty} S_{\text{out}}(\omega)\,\frac{\mathrm{d}\omega}{2\pi} \;=\; S_{\text{in}} \cdot \|H\|_{\mathcal{H}_2}^2, \tag{65}$$

where $\|H\|_{\mathcal{H}_2}^2 = \int |H(j\omega)|^2 \,\frac{\mathrm{d}\omega}{2\pi}$ is the squared $\mathcal{H}_2$ norm of the transfer function.

*Remark* 7.2 (Intuition: PSD as a frequency-by-frequency budget)*.* Think of any signal as a superposition of sine waves at different frequencies. The PSD $S(\omega)$ tells you how much power is carried by each frequency $\omega$—it is the signal's "energy budget" across frequencies.

A linear system cannot create new frequencies; it can only amplify or attenuate existing ones. The transfer function $H(j\omega)$ gives the complex gain at each frequency, so the output power at frequency $\omega$ is the input power times $|H(j\omega)|^2$. The total output variance is the sum (integral) over all frequencies.

The $\mathcal{H}_2$ norm $\|H\|_{\mathcal{H}_2}$ is therefore a single number that captures *how much total noise the system lets through.* A controller that minimises the $\mathcal{H}_2$ norm of the closed-loop transfer function is one that suppresses as much noise as possible.

*Remark* 7.3 (Gaussianity)*.* When the input is Gaussian and the system is linear, the output is also Gaussian. A Gaussian distribution is uniquely determined by its mean and variance. Therefore the transfer function $H(s)$, together with the input PSD, *completely determines the output probability distribution*—not merely its second moment.

## 7.3 Closed-Loop Transfer Functions

Under the proportional controller $U(s) = K\,E(s)$, the closed-loop becomes

$$s\,E(s) = -F_s\,K\,E(s) + W(s) \quad\Longrightarrow\quad E(s) = \frac{1}{s + F_s K}\,W(s). \tag{66}$$

Define the closed-loop transfer functions from the noise $W$ to each signal of interest:

**Noise to queue error.**

$$H_e(s) \;=\; \frac{E(s)}{W(s)} \;=\; \frac{1}{s + F_s K}. \tag{67}$$

**Noise to warp rate.** Since $U = K\,E$:

$$H_u(s) \;=\; \frac{U(s)}{W(s)} \;=\; \frac{K}{s + F_s K}. \tag{68}$$

**Noise to warp rate derivative.** Since $\dot{u} = K\,\dot{e}$, multiplication by $s$ in the Laplace domain gives

$$H_v(s) \;=\; s\,H_u(s) \;=\; \frac{K\,s}{s + F_s K}. \tag{69}$$

## 7.4 Output PSDs and Variances

Since $S_w = \sigma_w^2$ (constant), the output PSDs are:

**Queue error.**

$$S_e(\omega) \;=\; \frac{\sigma_w^2}{(F_s K)^2 + \omega^2}. \tag{70}$$

This is a Lorentzian with half-width $F_s K$.

The half-width $\omega_c = F_s K$ is the controller's *bandwidth*: it divides the frequency axis into two regimes.

- **Low frequencies** ($\omega \ll F_s K$): $S_e(\omega) \approx \sigma_w^2/(F_s K)^2$. The controller successfully suppresses slow queue variations—the error is small and nearly flat.
- **High frequencies** ($\omega \gg F_s K$): $S_e(\omega) \approx \sigma_w^2/\omega^2$. The controller is too slow to react, and the error behaves like uncontrolled integrated noise.

Increasing $K$ widens the bandwidth: the controller can fight faster disturbances, reducing the total queue error variance. But this comes at a cost, as we see next. The variance is

$$\mathrm{Var}[e] \;=\; \int_{-\infty}^{\infty} \frac{\sigma_w^2}{(F_s K)^2 + \omega^2}\,\frac{\mathrm{d}\omega}{2\pi} \;=\; \frac{\sigma_w^2}{2\,F_s\,K}, \tag{71}$$

recovering (56).

**Warp rate.**

$$S_u(\omega) \;=\; \frac{K^2\,\sigma_w^2}{(F_s K)^2 + \omega^2}. \tag{72}$$

The variance is

$$\mathrm{Var}[u] \;=\; \frac{K^2\,\sigma_w^2}{2\,F_s\,K} \;=\; \frac{K\,\sigma_w^2}{2\,F_s}, \tag{73}$$

recovering (57).

**Warp rate derivative.**

$$S_v(\omega) \;=\; \frac{K^2\,\omega^2\,\sigma_w^2}{(F_s K)^2 + \omega^2}. \tag{74}$$

The variance is

$$\mathrm{Var}[\dot{u}] \;=\; K^2\,\sigma_w^2 \int_{-\infty}^{\infty} \frac{\omega^2}{(F_s K)^2 + \omega^2}\,\frac{\mathrm{d}\omega}{2\pi} \;=\; +\infty. \tag{75}$$

The integral diverges because the integrand approaches $K^2\sigma_w^2$ as $|\omega| \to \infty$.

**Proposition 7.1** (Infinite warp roughness)**.** *Under the proportional controller* (53) *with the fluid model, the warp cost functional $J_u$ (Definition 4.1) is infinite almost surely. That is, the proportional controller produces warp trajectories that are nowhere differentiable.*

*Remark* 7.4 (Intuition: high-frequency passthrough). The transfer function $H_v(s) = Ks/(s + F_s K)$ is a *high-pass filter*. At low frequencies it attenuates (the controller changes $u$ slowly in response to slow queue drifts), but at high frequencies $|H_v(j\omega)| \to K$—it passes noise straight through with gain $K$, without any attenuation.

Since white noise has equal power at all frequencies, and $H_v$ does not roll off, the warp derivative picks up an infinite amount of high-frequency noise. This is the mathematical expression of the fact that "reacting proportionally to every instantaneous queue fluctuation produces jittery, rough control actions."

A controller that instead acts on $v = \dot{u}$ (controlling the rate of change of the warp) would introduce an additional $1/s$ factor, turning the high-pass into a band-pass that decays at high frequency, yielding finite $\mathrm{Var}[\dot{u}]$.

## 8 Second-Order Controller

The proportional controller of section 6 acts directly on the warp rate $u$, producing warp trajectories with infinite roughness (theorem 7.1). We now promote the control input to the *derivative* of the warp rate, $v = \dot{u}$, treating $u$ as an additional state variable.

### 8.1 Augmented State Space

The augmented system has state $\mathbf{x} = (e, u)^\top$ and control input $v = \dot{u}$. Absorbing the drift feedforward ($u$ represents the feedback component after subtracting $\delta$), the dynamics are

$$\mathrm{d}\begin{pmatrix} e \\ u \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & -F_s \\ 0 & 0 \end{pmatrix}}_{A} \begin{pmatrix} e \\ u \end{pmatrix} \mathrm{d}\tau + \underbrace{\begin{pmatrix} 0 \\ 1 \end{pmatrix}}_{B} v \; \mathrm{d}\tau + \underbrace{\begin{pmatrix} \sigma_w \\ 0 \end{pmatrix}}_{G} \mathrm{d}W. \tag{76}$$

The open-loop plant (from $v$ to $e$, without noise) is a *double integrator*: $v$ drives $u$, which in turn drives $e$ through the integrator $G(s) = 1/s$.

### 8.2 Control Law

**Definition 8.1** (Second-order controller)**.** The *second-order controller* sets

$$v(\tau) = K_1 \, e(\tau) - K_2 \, u(\tau), \tag{77}$$

where $K_1 > 0$ and $K_2 > 0$ are gains. The term $K_1 \, e$ provides a restoring force proportional to the queue error, and $-K_2 \, u$ provides damping proportional to the current warp rate.

*Remark* 8.1 (Physical analogy: damped harmonic oscillator). Under (77), the queue error satisfies the second-order ODE (ignoring noise):

$$\ddot{e} + K_2 \, \dot{e} + F_s \, K_1 \, e = 0. \tag{78}$$

This is a damped harmonic oscillator with natural frequency and damping ratio

$$\omega_n = \sqrt{F_s \, K_1}, \qquad \zeta = \frac{K_2}{2 \, \omega_n} = \frac{K_2}{2\sqrt{F_s \, K_1}}. \tag{79}$$

The queue error behaves like a mass on a spring, kicked by jitter noise, with $K_1$ controlling the spring stiffness (how fast it returns to centre) and $K_2$ controlling the damping (how quickly oscillations die out).

### 8.3 Closed-Loop Transfer Functions

In the Laplace domain, $V(s) = K_1 \, E(s) - K_2 \, U(s)$. With $s \, U = V$ (i.e. $U = V/s$), we have $(s + K_2) \, U = K_1 \, E$, so

$$U(s) = \frac{K_1}{s + K_2} \, E(s). \tag{80}$$

Substituting into the plant equation $s \, E = -F_s \, U + W$:

$$\left(s^2 + K_2 \, s + F_s \, K_1\right) E(s) = (s + K_2) \, W(s). \tag{81}$$

Denote the characteristic polynomial

$$D(s) = s^2 + K_2 \, s + F_s \, K_1 = s^2 + 2\zeta\omega_n \, s + \omega_n^2. \tag{82}$$

All roots lie in the open left half-plane for $K_1, K_2 > 0$, confirming stability.

The closed-loop transfer functions from the jitter noise $W$ to each output are:

**Noise to queue error.**

$$H_e(s) \;=\; \frac{s+K_2}{D(s)}. \tag{83}$$

**Noise to warp rate.**

$$H_u(s) \;=\; \frac{K_1}{D(s)}. \tag{84}$$

**Noise to warp rate derivative.**

$$H_v(s) \;=\; s\,H_u(s) \;=\; \frac{K_1\,s}{D(s)}. \tag{85}$$

This is a *band-pass filter*: it rolls off as $1/s$ at high frequency and vanishes at $s = 0$. Unlike the first-order case (eq. (69)), $|H_v(j\omega)| \to 0$ as $\omega \to \infty$.

## 8.4 Stationary Variances

We use the standard $\mathcal{H}_2$ norm formula for second-order systems. For a transfer function $H(s) = (n_1\,s + n_0)/(s^2 + a\,s + b)$ with $a, b > 0$:

$$\|H\|^2_{\mathcal{H}_2} \;=\; \frac{n_1^2\,b \;+\; n_0^2}{2\,a\,b}. \tag{86}$$

Applying this with $a = K_2$ and $b = F_s\,K_1$:

**Proposition 8.1** (Stationary variances, second-order controller)**.** *Under the second-order controller* (77) *with the fluid model, all three stationary variances are finite:*

$$\mathrm{Var}[e] \;=\; \sigma_w^2\,\frac{F_s\,K_1 + K_2^2}{2\,K_2\,F_s\,K_1}, \tag{87}$$

$$\mathrm{Var}[u] \;=\; \sigma_w^2\,\frac{K_1}{2\,K_2\,F_s}, \tag{88}$$

$$\mathrm{Var}[\dot{u}] \;=\; \sigma_w^2\,\frac{K_1^2}{2\,K_2}. \tag{89}$$

*Remark* 8.2 (Derivation). For example, $H_u(s) = K_1/(s^2 + K_2\,s + F_sK_1)$ has $n_1 = 0$, $n_0 = K_1$. Applying (86): $\|H_u\|^2 = (0 + K_1^2)/(2K_2F_sK_1) = K_1/(2K_2F_s)$. Multiplying by $\sigma_w^2$ gives (88). The other cases follow analogously.

*Remark* 8.3 (Comparison with first-order controller). The first-order controller (section 6.4) has $\mathrm{Var}[\dot{u}] = \infty$, so the warp cost $J_u$ is undefined. The second-order controller makes all three quantities finite and provides a two-parameter family $(K_1, K_2)$ of controllers. The Pareto surface in the three-dimensional space ($\mathrm{Var}[e]$, $\mathrm{Var}[u]$, $\mathrm{Var}[\dot{u}]$) is now a smooth manifold parameterised by $(K_1, K_2)$, allowing principled tradeoffs among queue accuracy, warp magnitude, and warp smoothness.

### 8.5 In Terms of Oscillator Parameters

Expressing the variances in terms of the natural frequency $\omega_n = \sqrt{F_s K_1}$ and damping ratio $\zeta = K_2/(2\omega_n)$:

$$\mathrm{Var}[e] = \frac{\sigma_w^2(1+4\zeta^2)}{4\,\zeta\,\omega_n}, \tag{90}$$

$$\mathrm{Var}[u] = \frac{\sigma_w^2\,\omega_n}{4\,\zeta\,F_s^2}, \tag{91}$$

$$\mathrm{Var}[\dot{u}] = \frac{\sigma_w^2\,\omega_n^3}{4\,\zeta\,F_s^2}. \tag{92}$$

*Remark* 8.4 (Role of $\omega_n$ and $\zeta$). The natural frequency $\omega_n$ controls the controller bandwidth: higher $\omega_n$ means faster response (lower Var$[e]$, but higher Var$[u]$ and Var$[\dot{u}]$). The damping ratio $\zeta$ controls oscillatory behaviour: underdamped ($\zeta < 1$) gives faster settling but overshoot; overdamped ($\zeta > 1$) gives slower, monotone convergence. All three variances scale as $1/\zeta$, and critically damped ($\zeta = 1$) is not necessarily optimal.

### 8.6 Dropping the Warp Magnitude Objective

The steady-state warp rate $u$ is dominated by the drift $\delta$, which is on the order of $10^{-5}$–$10^{-6}$ (a few ppm). Its stochastic variance Var$[u]$ is even smaller and always inaudible. What matters for audio quality is the *rate of change* $\dot{u}$: rapid pitch changes produce audible chirp artefacts, whereas a slowly-varying pitch offset is imperceptible.

Moreover, controlling Var$[\dot{u}]$ implicitly bounds Var$[u]$: if $\dot{u}$ is small, $u$ cannot wander far from its steady-state value.

We therefore optimise over two objectives only:

$$J(K_1, K_2) = \mathrm{Var}[e] + \lambda\ \mathrm{Var}[\dot{u}], \tag{93}$$

where $\lambda > 0$ is a design parameter weighting queue accuracy against warp smoothness.

### 8.7 Optimal Gains

Substituting (87) and (89) into (93):

$$J = \sigma_w^2\left[\frac{1}{2\,K_2} + \frac{K_2}{2\,F_s\,K_1} + \frac{\lambda\,K_1^2}{2\,K_2}\right]. \tag{94}$$

Setting $\partial J/\partial K_1 = 0$:

$$-\frac{K_2}{2\,F_s\,K_1^2} + \frac{\lambda\,K_1}{K_2} = 0 \implies K_2^2 = 2\,\lambda\,F_s\,K_1^3. \tag{95}$$

Setting $\partial J/\partial K_2 = 0$:

$$-\frac{1}{2\,K_2^2} + \frac{1}{2\,F_s\,K_1} - \frac{\lambda\,K_1^2}{2\,K_2^2} = 0 \implies K_2^2 = F_s\,K_1\,(1+\lambda\,K_1^2). \tag{96}$$

Equating (95) and (96):

$$2\,\lambda\,F_s\,K_1^3 = F_s\,K_1 + \lambda\,F_s\,K_1^3 \implies \lambda\,K_1^2 = 1, \tag{97}$$

giving the optimal gains:

**Proposition 8.2** (Optimal second-order gains)**.** *The cost* $J = \mathrm{Var}[e] + \lambda\ \mathrm{Var}[\dot{u}]$ *is minimised by*

$$K_1^* \;=\; \frac{1}{\sqrt{\lambda}}, \qquad K_2^* \;=\; \sqrt{2\,F_s\,K_1^*} \;=\; \sqrt{\frac{2\,F_s}{\lambda^{1/2}}}. \tag{98}$$

*The corresponding oscillator parameters are*

$$\omega_n^* \;=\; \sqrt{F_s\,K_1^*} \;=\; \left(\frac{F_s^2}{\lambda}\right)^{1/4}, \qquad \boxed{\zeta^* \;=\; \frac{1}{\sqrt{2}} \;\approx\; 0.707.} \tag{99}$$

*Remark* 8.5 (Universal damping ratio). The optimal damping ratio $\zeta^* = 1/\sqrt{2}$ is **independent of** $\lambda$, the jitter intensity $\sigma_w$, and the sample rate $F_s$. Only the natural frequency $\omega_n$ changes with the design parameters. The value $\zeta = 1/\sqrt{2}$ is the Butterworth (maximally flat magnitude) damping, well known in filter design and classical optimal control.

## 8.8 Optimal Variances

Substituting the optimal gains into the variance expressions:

$$\mathrm{Var}[e]^* \;=\; \frac{3\,\sigma_w^2}{2\sqrt{2}\,\omega_n^*}, \tag{100}$$

$$\mathrm{Var}[\dot{u}]^* \;=\; \frac{\sigma_w^2\,(\omega_n^*)^3}{2\sqrt{2}\,F_s^2}, \tag{101}$$

$$J^* \;=\; \mathrm{Var}[e]^* + \lambda\ \mathrm{Var}[\dot{u}]^* \;=\; \frac{3\,\sigma_w^2}{2\sqrt{2}\,\omega_n^*} + \frac{\sigma_w^2\,(\omega_n^*)^3}{2\sqrt{2}\,F_s^2}\cdot\lambda. \tag{102}$$

Since $\omega_n^* = (F_s^2/\lambda)^{1/4}$, both terms depend on $\lambda$ through $\omega_n^*$ alone: increasing $\lambda$ (more weight on smoothness) decreases $\omega_n^*$, widening $\mathrm{Var}[e]$ and shrinking $\mathrm{Var}[\dot{u}]$.

## 8.9 Steady-State Bias

The second-order controller has a nonzero steady-state queue error in the presence of constant clock drift. This section analyses the bias, explains its root cause in terms of system type, and outlines a path to eliminating it.

### 8.9.1 Equilibrium Analysis

In steady state all derivatives vanish. From the plant equation (49) (deterministic part):

$$\frac{\mathrm{d}e}{\mathrm{d}\tau} = 0 \quad\Longrightarrow\quad F_s\,(\delta - u_{\mathrm{ss}}) = 0 \quad\Longrightarrow\quad u_{\mathrm{ss}} = \delta. \tag{103}$$

From the controller law (**??**):

$$\frac{\mathrm{d}u}{\mathrm{d}\tau} = 0 \quad\Longrightarrow\quad K_1\,e_{\mathrm{ss}} = K_2\,u_{\mathrm{ss}} = K_2\,\delta \quad\Longrightarrow\quad \boxed{e_{\mathrm{ss}} = \frac{K_2}{K_1}\,\delta.} \tag{104}$$

The bias is proportional to the drift $\delta$ and to the ratio $K_2/K_1$.

**Numerical magnitude.** For the default gains $K_1 = 10^{-5}$ and $K_2 = \sqrt{2\,F_s\,K_1} \approx 0.98$ at $F_s = 48\,000$:

$$\frac{K_2}{K_1} \approx 9.8 \times 10^4. \tag{105}$$

A drift of $\delta = 10$ ppm $= 10^{-5}$ gives a steady-state bias of

$$e_{\mathrm{ss}} \approx 0.98 \text{ samples } \approx\; 20\ \mu\text{s at } 48\,\text{kHz}, \tag{106}$$

which is negligible compared to jitter-induced fluctuations. For larger drifts (e.g. $\delta = 100$ ppm), the bias grows to $\approx 10$ samples $\approx 200\ \mu$s.

### 8.9.2 Root Cause: System Type

The open-loop transfer function from error $e$ to the plant input $u$ contains exactly *one* integrator (the warp state $u$), but the damping term $-K_2\,u$ makes it a *leaky* integrator:

$$U(s) = \frac{K_1}{s + K_2}\,E(s). \tag{107}$$

In classical control terminology, this is a **Type 1** system. A Type 1 system rejects step disturbances with zero steady-state error but has nonzero error for a ramp. The constant drift $\delta$ enters the plant as a step in $\dot{e}$, which from the perspective of the error $e$ is a ramp input—hence the nonzero position error.

### 8.9.3 Why the H2 Cost Misses the Bias

The H2 optimisation (Section 8.6) minimises

$$J = \mathrm{Var}[e] + \lambda\ \mathrm{Var}[\dot{u}], \tag{108}$$

which captures only the *variance* (fluctuation around the mean), not the *mean* itself. The steady-state bias $e_{\mathrm{ss}} = (K_2/K_1)\,\delta$ is deterministic and does not appear in the variance. Consequently, the H2-optimal gains make no attempt to reduce the bias, focusing entirely on minimising the stochastic fluctuations.

### 8.9.4 Towards Zero Bias: Type 2 Design

To achieve $e_{\mathrm{ss}} = 0$ for constant drift, one needs a **Type 2** system—two integrators in the open loop. This requires adding a third state variable that accumulates the integral of the error:

$$\frac{\mathrm{d}\xi}{\mathrm{d}\tau} = e, \tag{109}$$

$$\frac{\mathrm{d}u}{\mathrm{d}\tau} = K_1\,e + K_0\,\xi - K_2\,u, \tag{110}$$

where $\xi$ is the integrated error and $K_0 > 0$ is the integral gain. The controller now has three gains: $K_0$ (integral), $K_1$ (proportional), and $K_2$ (damping).

**Steady-state verification.** In steady state, $\dot{\xi} = 0$ implies $e_{\mathrm{ss}} = 0$ immediately—regardless of the drift $\delta$ or the gain values. The integrator state settles to $\xi_{\mathrm{ss}} = K_2\,\delta/K_0$, and the warp state to $u_{\mathrm{ss}} = \delta$ as before.

#### 8.9.5 PID Interpretation

The control law (110) is a PID controller applied to the queue error:

- $K_0\,\xi = K_0\!\int\! e \, \mathrm{d}\tau$: **Integral** action (eliminates bias),
- $K_1\, e$: **Proportional** action (restoring force),
- $-K_2\, u$: **Derivative** action on the warp state (damping).

The previous second-order controller is the PD subset ($K_0 = 0$).

#### 8.9.6 Closed-Loop Transfer Functions

The state-space representation with state $\mathbf{x} = (e, \xi, u)^\top$ and noise input $w$:

$$\underbrace{\begin{pmatrix} \dot{e} \\ \dot{\xi} \\ \dot{u} \end{pmatrix}}_{\dot{\mathbf{x}}} = \underbrace{\begin{pmatrix} 0 & 0 & -F_s \\ 1 & 0 & 0 \\ K_1 & K_0 & -K_2 \end{pmatrix}}_{A} \begin{pmatrix} e \\ \xi \\ u \end{pmatrix} + \underbrace{\begin{pmatrix} \sigma_w \\ 0 \\ 0 \end{pmatrix}}_{B} \dot{W}. \tag{111}$$

The closed-loop characteristic polynomial is

$$p(s) = s^3 + K_2\, s^2 + F_s\, K_1\, s + F_s\, K_0. \tag{112}$$

The transfer function from noise $w$ to queue error $e$ is

$$H_e(s) = \frac{\sigma_w\, s\,(s + K_2)}{p(s)}, \tag{113}$$

and from noise to warp derivative $\dot{u}$:

$$H_{\dot{u}}(s) = \frac{\sigma_w\, s\,(K_1\, s + K_0)}{p(s)}. \tag{114}$$

Note the *extra zero at* $s = 0$ in both transfer functions compared to the PD controller. This zero suppresses the low-frequency response, which is precisely the mechanism by which integral action rejects the DC disturbance (drift).

#### 8.9.7 Dominant-Pole Gain Design

Rather than solving the full 3-parameter H2 optimisation (which yields transcendental equations), we exploit the separation of timescales. Choose the integral gain $K_0$ small enough that the characteristic polynomial factors approximately as

$$p(s) \;\approx\; (s^2 + 2\zeta\omega_n\, s + \omega_n^2)\,(s + p_0), \tag{115}$$

where $p_0 \ll \zeta\omega_n$ is a slow real pole governing the integral mode. Expanding and matching with (112):

$$\begin{aligned} K_2 &= 2\zeta\omega_n + p_0 \;\approx\; 2\zeta\omega_n, && (116)\\ F_s\, K_1 &= \omega_n^2 + 2\zeta\omega_n\, p_0 \;\approx\; \omega_n^2, && (117)\\ F_s\, K_0 &= \omega_n^2\, p_0 \quad \Longrightarrow \quad \boxed{K_0 = K_1\, p_0.} && (118) \end{aligned}$$

**Proposition 8.3** (PID gain design). *Keep the Butterworth-optimal PD gains (Proposition 8.2):*

$$K_1 = \frac{1}{\sqrt{\lambda}}, \qquad K_2 = \sqrt{2\,F_s\,K_1}, \qquad \zeta = \frac{1}{\sqrt{2}}. \tag{119}$$

*Set the integral gain via its settling time $T_I$:*

$$p_0 = \frac{1}{T_I}, \qquad K_0 = \frac{K_1}{T_I}. \tag{120}$$

*The integral mode eliminates the steady-state bias with time constant $T_I$, while the dominant oscillator dynamics are essentially unchanged.*

*Remark* 8.6 (Stability margin). The Routh condition requires $K_0 < K_1\,K_2$. Since $K_0 = K_1/T_I$ and $K_2 \approx \omega_n\sqrt{2}$, the condition becomes $T_I > 1/K_2 \approx 1/\omega_n$. For $K_1 = 10^{-5}$ and $F_s = 48\,000$: $\omega_n \approx 0.69$, so $T_I > 1.4$ s. A practical choice of $T_I = 30$ s gives a large stability margin.

### 8.9.8 Effect on Variances

Solving the Lyapunov equation $AP + PA^\top + BB^\top = 0$ for the PID state-space representation yields the exact stationary variance:

$$\mathrm{Var}[e]_{\mathrm{PID}} = \frac{\sigma_w^2\,(K_2^2 + F_s\,K_1)}{2\,F_s\,(K_1\,K_2 - K_0)}. \tag{121}$$

Comparing with the PD result ($K_0 = 0$):

$$\frac{\mathrm{Var}[e]_{\mathrm{PID}}}{\mathrm{Var}[e]_{\mathrm{PD}}} = \frac{K_1\,K_2}{K_1\,K_2 - K_0} = \frac{1}{1 - K_0/(K_1\,K_2)}. \tag{122}$$

Since $K_0 > 0$, the PID variance is *slightly larger* than the PD variance. The integral mode adds a slow resonance that admits more low-frequency noise energy.

**Numerical overhead.** For $K_0 = K_1/T_I$ with $T_I = 30$ s and $K_2 \approx 0.98$:

$$\frac{K_0}{K_1\,K_2} = \frac{1}{K_2\,T_I} \approx \frac{1}{0.98 \times 30} \approx 3.4\%. \tag{123}$$

The variance overhead is thus $\approx 3.5\%$—negligible compared to the benefit of eliminating steady-state bias entirely.

### 8.9.9 The Full Three-Objective Cost

The complete cost function incorporating all three objectives is

$$J = \underbrace{\mathbb{E}[e]^2}_{J_1\text{ (bias)}} + \underbrace{\mathrm{Var}[e]}_{J_2\text{ (regulation)}} + \lambda \underbrace{\mathrm{Var}[\dot{u}]}_{J_3\text{ (smoothness)}}. \tag{124}$$

For the PID controller:

$$J_1 = 0 \qquad \text{(structurally zero for any } K_0 > 0\text{)}, \tag{125}$$

$$J_2 = \frac{\sigma_w^2\,(K_2^2 + F_s\,K_1)}{2\,F_s\,(K_1\,K_2 - K_0)} \qquad \text{(increases with } K_0\text{)}, \tag{126}$$

$$J_3 \approx \frac{\sigma_w^2\,\omega_n^3}{2\sqrt{2}\,F_s^2} \qquad \text{(weakly affected)}. \tag{127}$$

*Remark* 8.7 (Optimal $K_0$ depends on unknown drift). Since $J_1 = 0$ for any $K_0 > 0$ but $J_2$ increases with $K_0$, the H2-optimal integral gain is $K_0 \to 0^+$—as small as possible while still nonzero. In practice, $K_0$ must be large enough that the integral mode settles within a reasonable time (e.g. $T_I = 30\,\mathrm{s}$), giving the dominant-pole design (120) as the pragmatic optimum.

The user controls the $J_2$–$J_3$ tradeoff through $K_1$ (equivalently $\lambda$ or $\omega_n$), while $K_0 = K_1/T_I$ handles the bias independently.

### 8.10 Discrete-Time Implementation

In practice the controller runs at discrete epochs of duration $h$ (the scaling interval). At each epoch $k$:

1. Observe the queue error $e_k = q_k - q^*$.
2. Update the integral state:
$$\xi_{k+1} \;=\; \xi_k \;+\; e_k\, h. \tag{128}$$
3. Compute the control acceleration:
$$v_k \;=\; K_1\, e_k \;+\; K_0\, \xi_k \;-\; K_2\, u_k. \tag{129}$$
4. Update the warp state by Euler integration:
$$u_{k+1} \;=\; u_k \;+\; v_k\, h. \tag{130}$$
5. Clamp to safety bounds and set the resampler coefficient:
$$\texttt{freq\_coeff}_k \;=\; 1 + \mathrm{clamp}(u_k,\; -u_{\max},\; u_{\max}). \tag{131}$$

*Remark* 8.8 (Anti-windup). When the warp state $u$ saturates at $\pm u_{\max}$, the integrator $\xi$ must be frozen to prevent windup. In the implementation, $\xi$ is not updated when $u$ is clamped and the error has the same sign as the saturation direction. This ensures the integral state does not grow unboundedly during transients or fault conditions.

*Remark* 8.9 (No feedforward drift estimate). An external drift estimate $\hat{\delta}$ (e.g. from RTP timestamp ratios) is not required. The integral action drives the steady-state bias to exactly zero for any constant drift, making feedforward unnecessary. Omitting feedforward avoids step disturbances from estimator artefacts (NTP corrections, session restarts, bias in early samples).

## 9 Skewness and Asymmetric Control

The preceding analysis assumes symmetric Gaussian noise, yielding a symmetric stationary distribution for the queue error. In practice, network jitter is *one-sided*: packets can be delayed but never arrive before the minimum propagation time. This produces a skewed queue error distribution whose tails are asymmetric, with direct consequences for the underrun probability.

This section develops a Fokker–Planck analysis of the queue error under asymmetric feedback, yielding a closed-form stationary density and exact skewness control.

## 9.1 Physical Origin of Skewness

Each packet's transit delay decomposes as $d_k = d_{\min} + j_k$ where $j_k \geq 0$. The non-negative constraint on $j_k$ is physical (speed of light plus serialization) and implies that per-packet jitter has strictly positive skewness: $\kappa_3[j_k] > 0$. Common models include exponential jitter ($\gamma_3 = 2$) and gamma jitter ($\gamma_3 = 2/\sqrt{\alpha}$).

The aggregate noise $w_k$ driving the queue in each control epoch inherits this skewness. By the CLT with Edgeworth correction, if $N$ packets arrive per epoch:

$$\gamma_3[w_k] = \frac{\gamma_3[j]}{\sqrt{N}}, \tag{132}$$

which shrinks as $1/\sqrt{N}$ but remains nonzero for finite $N$.

Through the linear controller, the queue error skewness is

$$\kappa_3[e] = \kappa_3[w] \cdot \int_0^\infty h_e(\tau)^3 \, \mathrm{d}\tau, \tag{133}$$

where $h_e$ is the closed-loop impulse response from noise to error. For a stable linear system with all-positive impulse response, $\int h_e^3 > 0$, so the output skewness has the same sign as the input.

*Remark* 9.1 (Linear controllers cannot eliminate skewness). Skewness is a third-order cumulant. A linear filter's transfer function controls the power spectrum (second-order), not the bispectrum (third-order). To set $\kappa_3[e] = 0$ from nonzero input skewness requires $\int h_e^3 = 0$, which for the class of stable minimum-phase responses needed here cannot be achieved. Independent control of skewness requires a *nonlinear* controller element.

## 9.2 Asymmetric Proportional Controller

The simplest nonlinear element that targets skewness is an *asymmetric gain*: the controller responds more aggressively when the queue error is in the dangerous (underrun) direction.

Replace the constant proportional gain $K$ with a piecewise gain:

$$K(e) \;=\; \begin{cases} K_+ & \text{if } e \geq 0 \quad \text{(queue above target)}, \\ K_- & \text{if } e < 0 \quad \text{(queue below target)}, \end{cases} \tag{134}$$

with $K_- > K_+$ to respond more aggressively in the underrun direction.

For analytical tractability, we analyse the first-order (proportional) controller $u = K(e)\, e$ under the fluid approximation. The queue error SDE becomes a *piecewise-linear* diffusion:

$$\mathrm{d}e \;=\; -F_s\, K(e)\, e \;\mathrm{d}\tau \;+\; \sigma_w \;\mathrm{d}W. \tag{135}$$

This is an Ornstein–Uhlenbeck process with rate $\alpha_+ = F_s\, K_+$ for $e > 0$ and rate $\alpha_- = F_s\, K_-$ for $e < 0$.

## 9.3 Fokker–Planck Equation

Write $D = \sigma_w^2/2$ for the diffusion coefficient. The stationary Fokker–Planck equation for the probability density $p(e)$ is

$$\frac{\partial}{\partial e}\big[\alpha(e)\, e\, p(e)\big] + D\,\frac{\partial^2 p}{\partial e^2} = 0, \tag{136}$$

where $\alpha(e) = \alpha_\pm$ on each half-line. On each half, (136) is the standard Fokker–Planck equation for an OU process, whose stationary solution is Gaussian:

$$p(e) \;=\; \begin{cases} C\,\exp\!\Big(-\dfrac{e^2}{2\sigma_+^2}\Big) & e \geq 0, \\ C\,\exp\!\Big(-\dfrac{e^2}{2\sigma_-^2}\Big) & e < 0, \end{cases} \tag{137}$$

with half-variances

$$\sigma_+^2 = \frac{D}{\alpha_+} = \frac{\sigma_w^2}{2\,F_s\,K_+}, \qquad \sigma_-^2 = \frac{D}{\alpha_-} = \frac{\sigma_w^2}{2\,F_s\,K_-}. \tag{138}$$

**Matching conditions.** At $e = 0$: (i) continuity of $p$ is ensured by using the same constant $C$ on both halves; (ii) the probability current $J = -\alpha\, e\, p - D\, p'$ vanishes identically on each half (since $p$ satisfies the OU equation), ensuring zero-current continuity.

**Normalisation.**

$$1 = C\int_0^\infty e^{-e^2/(2\sigma_+^2)}\,\mathrm{d}e + C\int_{-\infty}^0 e^{-e^2/(2\sigma_-^2)}\,\mathrm{d}e = C\,\sqrt{\tfrac{\pi}{2}}\,(\sigma_+ + \sigma_-) \tag{139}$$

gives $C = \sqrt{2/\pi}\,/\,(\sigma_+ + \sigma_-)$.

**Definition 9.1** (Split normal distribution)**.** The density (137) is a *split normal* (two-piece Gaussian) with scale parameters $\sigma_+$ and $\sigma_-$. When $\sigma_+ = \sigma_-$ it reduces to the ordinary Gaussian.

## 9.4 Moments of the Split Normal

Let $r = \sigma_-/\sigma_+ = \sqrt{K_+/K_-}$ be the asymmetry ratio ($r < 1$ when the underrun side is tighter). All moments are expressible in closed form.

**Mean.**

$$\mathbb{E}[e] = \sqrt{\frac{2}{\pi}}\,(\sigma_+ - \sigma_-) = \sqrt{\frac{2}{\pi}}\,\sigma_+\,(1 - r), \tag{140}$$

which is *positive* when $K_- > K_+$ ($r < 1$): the asymmetric controller shifts the mean toward the safe (positive) side.

**Second moment and variance.**

$$\mathbb{E}[e^2] = \frac{\sigma_+^3 + \sigma_-^3}{\sigma_+ + \sigma_-} = \sigma_+^2\,\frac{1 + r^3}{1 + r}, \tag{141}$$

$$\mathrm{Var}[e] = \sigma_+^2\left[\frac{1 + r^3}{1 + r} - \frac{2}{\pi}(1 - r)^2\right]. \tag{142}$$

**Skewness.** For small asymmetry $r = 1 - \epsilon$ with $\epsilon \ll 1$, the skewness coefficient simplifies to:

$$\gamma_3 \;\approx\; \sqrt{\frac{2}{\pi}}\;\frac{(4/\pi - 1)\,\epsilon}{\big(1 - 2/\pi\big)^{3/2}} \;\approx\; 0.31\,\epsilon. \tag{143}$$

*Remark* 9.2 (Interpretation). Each unit of gain asymmetry ($\epsilon = 1 - r = 1 - \sqrt{K_+/K_-}$) produces $\approx 0.31$ units of skewness. Since $\gamma_3$ of the input noise is typically 0.3–1.0, a modest asymmetry of $K_-/K_+ \approx 1.5$ ($\epsilon \approx 0.18$, $\gamma_3 \approx 0.06$) can already partially compensate the intrinsic jitter skewness.

### 9.5 Underrun Probability

The probability that the queue error exceeds a threshold $-M$ (i.e. the queue drops $M$ samples below target) is the left tail of the split normal:

$$P(e < -M) = \frac{\sigma_-}{\sigma_+ + \sigma_-}\ \operatorname{erfc}\left(\frac{M}{\sigma_- \sqrt{2}}\right). \tag{144}$$

For the symmetric controller ($\sigma_+ = \sigma_- = \sigma$):

$$P_{\text{sym}}(e < -M) = \tfrac{1}{2}\ \operatorname{erfc}\big(M/(\sigma\sqrt{2})\big). \tag{145}$$

The key observation is that in the asymmetric case, the argument of the erfc is $M/\sigma_-$, and since $\sigma_- < \sigma_+$ (the underrun side is tighter), the effective number of standard deviations *increases*:

$$\frac{M}{\sigma_-} = \frac{M}{\sigma_+ r} = \frac{1}{r}\frac{M}{\sigma_+}. \tag{146}$$

**Proposition 9.1** (Underrun reduction)**.** *With asymmetry ratio $r = \sqrt{K_+/K_-}$, a buffer margin of $M$ provides $M/(\sigma_+ r)$ effective standard deviations of protection on the underrun side, compared to $M/\sigma$ for the symmetric case. For $K_-/K_+ = 2$ ($r = 1/\sqrt{2}$):*

- *A $3\sigma$ margin becomes effectively $4.2\sigma$,*
- *The underrun probability drops by a factor of $\approx 10$.*

### 9.6 The Variance–Skewness Tradeoff

The asymmetric controller shifts the mean to $\mathbb{E}[e] > 0$ (the queue sits slightly above target on average). This is a deliberate asymmetric operating point that protects the underrun tail.

For small asymmetry, the variance penalty is *second-order* while the underrun benefit is *first-order*:

**Proposition 9.2** (Variance–skewness tradeoff)**.** *At leading order in $\epsilon = 1 - r$:*

- *Underrun probability decreases exponentially (via the increased $M/\sigma_-$ ratio),*
- *Variance increases as $1 + O(\epsilon^2)$ (quadratic cost),*
- *Mean shift is $O(\epsilon)$ (linear, compensated by integral action if present).*

*The tradeoff strongly favours small asymmetry when the buffer margin $M$ is large relative to $\sigma$.*

### 9.7 Extension to the PID Controller

The Fokker–Planck analysis above uses a first-order (proportional) controller for analytical tractability. For the PID controller (Section 8.9), the state space is three-dimensional $(e, \xi, u)$ and the Fokker–Planck equation becomes a 3D PDE, which does not admit closed-form solutions in general.

However, the qualitative results carry over:

1. The marginal density of $e$ remains approximately split-normal when the gain asymmetry is a slowly-varying modulation of the linear dynamics (*adiabatic approximation*).
2. The integral action ($K_0$) compensates the mean shift introduced by the asymmetric gain, restoring $\mathbb{E}[e] = 0$ while preserving the asymmetric tail structure.

3. The dominant pole separation ($p_0 \ll \omega_n$) means the integral mode operates on a much slower timescale and does not interact with the asymmetric gain dynamics.

In practice, the asymmetric PID controller would use:

$$v \;=\; K_1(e)\, e + K_0\, \xi - K_2\, u, \qquad K_1(e) = K_1\,(1 + \beta\; \mathrm{sgn}(-e)), \tag{147}$$

where $\beta \in [0, 1)$ controls the asymmetry. The effective gains are $K_- = K_1(1 + \beta)$ for $e < 0$ and $K_+ = K_1(1 - \beta)$ for $e > 0$, giving $r = \sqrt{(1 - \beta)/(1 + \beta)}$.